# NONVANISHING OF DEGREE N COHOMOLOGY IS INHERITED ALONG A LIMIT OF FINITE ETALE COVERS OF DEGREE PRIME TO L

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The proof is the author's complete argument for both implications, and it exhibits the transfers named in the last sentence of the statement.

*Editorial note.* Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

*Editorial note.* The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

*Editorial note.* The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

*Allowed standard prerequisites (editorial).*

- Limit of a directed system of finite etale covers of degree prime to l. Same chapter, the description of the limit X of a directed system of finite etale S-schemes of constant degree prime to l and the functors g^{-1}, g_* attached to the projection g : X -> S. Supplies the morphism g and the two transfers used in the proof.
- Behaviour of l-power torsion cohomology under finite etale covers of degree prime to l. Same chapter, the results stating that a finite etale morphism of degree prime to l induces an injection on l-power torsion cohomology, and the passage to the limit of such covers. Supplies both implications, since transfer and restriction compose to multiplication by the degree.

*Source reference.* Original theorem 09YQ: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 7400–7400. Navigation pointer invoked at proof line 10567; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original remark 03SH: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 10330–10330. Navigation pointer invoked at proof

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line 10538; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original proposition 03QP: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 8245–8245. Navigation pointer invoked at proof line 10589; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original lemma 09Z1: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 7535–7535. Navigation pointer invoked at proof line 10590; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original proposition 03QC: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 8176–8176. Navigation pointer invoked at proof line 10594; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

**Lemma 0.1.** *Let $\ell$ be a prime number and $n$ an integer $> 0$. Let $S$ be a quasi-compact and quasi-separated scheme. Let $X = \lim_{i \in I} X_i$ be the limit of a directed system of $S$-schemes each $X_i \to S$ being finite étale of constant degree relatively prime to $\ell$. The following are equivalent:*

1. *there exists an $\ell$-power torsion sheaf $\mathcal{G}$ on $S$ such that $H^n_{étale}(S, \mathcal{G}) \neq 0$ and*
2. *there exists an $\ell$-power torsion sheaf $\mathcal{F}$ on $X$ such that $H^n_{étale}(X, \mathcal{F}) \neq 0$.*

*In fact, given $\mathcal{G}$ we can take $\mathcal{F} = g^{-1}\mathcal{F}$ and given $\mathcal{F}$ we can take $\mathcal{G} = g_*\mathcal{F}$.*

*Proof.* Let $g : X \to S$ and $g_i : X_i \to S$ denote the structure morphisms. Fix an $\ell$-power torsion sheaf $\mathcal{G}$ on $S$ with $H^n_{étale}(S, \mathcal{G}) \neq 0$. The system given by $\mathcal{G}_i = g_i^{-1}\mathcal{G}$ satisfy the conditions of Theorem 09YQ with colimit sheaf given by $g^{-1}\mathcal{G}$. This tells us that:

$$\operatorname{colim}_{i \in I} H^n_{étale}(X_i, g_i^{-1}\mathcal{G}) = H^n_{étale}(X, \mathcal{G})$$

By virtue of the $g_i$ being finite étale morphism of degree prime to $\ell$ we can apply "la méthode de la trace" and we find the maps

$$H^n_{étale}(S, \mathcal{G}) \to H^n_{étale}(X_i, g_i^{-1}\mathcal{G})$$

are all injective (and compatible with the transition maps). See Section 03SH. Thus, the colimit is non-zero, i.e., $H^n(X, g^{-1}\mathcal{G}) \neq 0$, giving us the desired result with $\mathcal{F} = g^{-1}\mathcal{G}$.

Conversely, suppose given an $\ell$-power torsion sheaf $\mathcal{F}$ on $X$ with $H^n_{étale}(X, \mathcal{F}) \neq 0$. We note that since the $g_i$ are finite morphisms the higher direct images vanish (Proposition 03QP). Then, by applying Lemma 09Z1 we may also conclude the same for $g$. The vanishing of the higher direct images tells us that $H^n_{étale}(X, \mathcal{F}) = H^n(S, g_*\mathcal{F}) \neq 0$ by Leray (Proposition 03QC) giving us what we want with $\mathcal{G} = g_*\mathcal{F}$. □